

2. Mass Movements: It is the downslope movement of soil, rock, and other debris under the influence of gravity. Weathering is not a prerequisite.

Types:

1. Rapid: Landslides (due to deforestation), avalanche, earthflow, mudflow (leads to soil liquefaction).
2. Slow: Creep (slow down slope movement of particles), solifluction (slow progressive movement of mass down a slope seen mainly in temperate regions).

Erosion: It involves transportation of rock debris through geomorphic agents like running water, groundwater, glaciers, winds and waves.



LANDFORMS

- Landforms are natural or human-made features on the Earth's surface. They include hills, mountains, plains, and more. The landforms are formed due to various agents such as river, glaciers, wind, sea waves, groundwater, etc.

Types of Landforms

- Erosional Landforms:** They are created by the process of erosion.

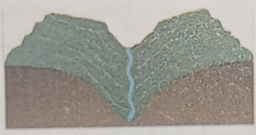
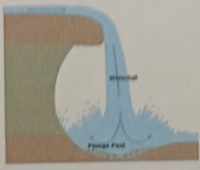
- Depositional Landforms:** These are created when sediments are deposited in areas of reduced flow energy.

Landforms Created by River

- A river goes through three stages: *Youthful, mature, and old*. The course of a river includes *the upper stage, the middle stage, and the lower stage*.

Youthful Stage	Mature Stage	Old Stage
River flows with high <i>speed</i> . Eg. V-shaped valley, Gorges, Canyon, Waterfalls, Entrenched meander	The river flows from <i>high gradient to flat plain</i> . Eg. Meanders	The final stage of river's journey Eg. Ox-bow lakes, Delta, Flood plains

Erosional Landforms:

Landforms	Definitions	Pictures
V-shaped Valley	Narrow valley with steep sides formed by vertical river erosion.	
Gorge	Deep, narrow valley with very steep sides. Commonly developed in hard, resistant rocks.	
Canyon	Large, deep gorge with layered rock sides, commonly form in horizontal sedimentary rock layers.	
Plunge pool	Deep hole created at the base of a plunge waterfall.	

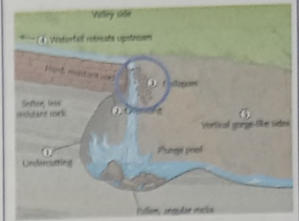
Peneplain

A nearly level plain formed due to prolonged erosion, reducing the land almost to its base level.



Waterfall

A sudden vertical drop in a river, mainly due to the erosion of soft rock, which creates a steep fall.



Cataract Waterfall

When waterfall forms where hard rock and soft rock create an uneven river bed, causing heavy, turbulent flow.



Cascade Waterfall

A waterfall that consists of series of small falls where water flows over rocks or boulders (multi-step waterfalls).



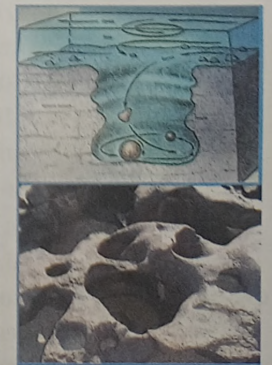
Plunge Waterfall

A waterfall where water falls straight down without touching the cliff face below. (Water velocity is very high).

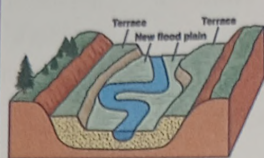


Potholes

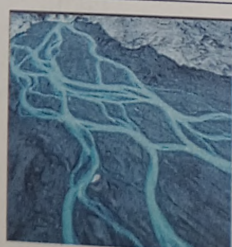
Round holes in the river bed formed by swirling pebbles. (The process of wearing away rocks by friction of particles carried by a river is called Abrasion).

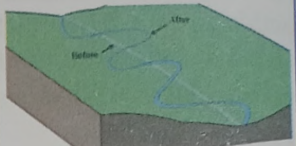
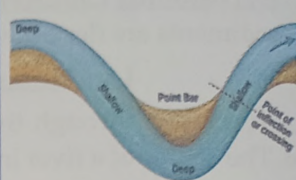
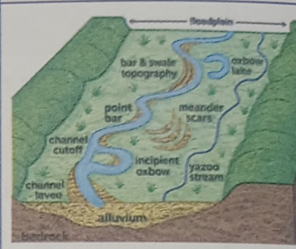
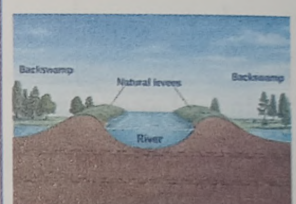


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Rapids	Fast, turbulent rivers flow horizontally over alternating hard rocks (resilient rock) and soft rocks (less resilient rock) creating step-like structure.	
River terraces	Flat steps along a river valley that were once floodplains (areas where floods spread), but got left at a higher level when the river cut deeper into the land.	
Entrenched Meander	Generally found in mountainous regions, where a river cuts deep into rock when uplifted land blocks its path, forming steep-sided meanders.	

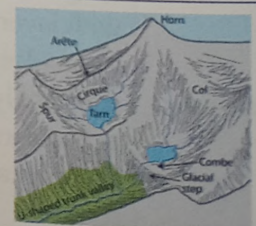
• Depositional Landforms:

Landforms	Definitions	Pictures
Alluvial Fans	A fan-shaped deposit of sediments formed where a river slows down suddenly on leaving a mountainous region.	
Braided Channels	A network of interweaving river channels formed due to heavy sediment deposition within the river.	
Ox-Bow Lake	A crescent shaped lake formed from a cut-off meander due to deposition sealing the ends.	

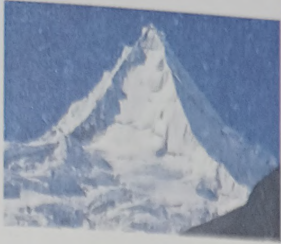
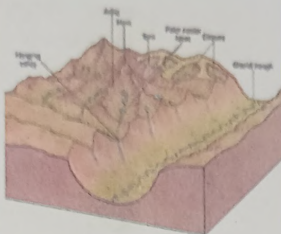
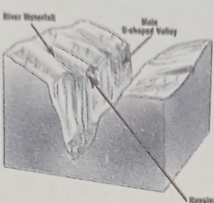
Meanders	Bends or loops in a river formed due to erosion on the outer bank (concave bank) and deposition on the inner bank (convex bank/point bar).	
Point Bars	Deposition of sediments on the inner side (convex bank) of a meander bend due to slower flow of water.	
Flood Plains	Broad, flat areas along a river, formed by the deposition of fine sediments (alluvium) during repeated floods.	
Levees	Raised embankments along river banks formed by deposition of heavier sediments during floods.	

Landforms Created by Glaciers

• Erosional Landforms:

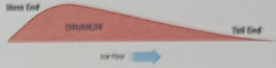
Landforms	Definitions	Pictures
Cirque	A bowl-shaped, steep-sided hollow formed at the head of a glacier due to erosion.	
Tarn Lake	A small lake formed when water fills up in a cirque.	
Arete (Ridges)	A sharp, narrow ridge formed between two adjacent cirques due to glacial erosion.	

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Horn	A pointed, pyramid-like peak formed when three or more cirques erode a mountain from different sides.	
U shaped/ Glacial Valley	A wide, flat-bottomed "U" shaped valley with steep sides formed by glacier movement. (unlike narrow V-shaped river valleys).	
Hanging Valley	A smaller valley left above the main U-shaped glacial valley (often giving rise to waterfalls).	
Fjord	When sea water enters into the glacial valley.	

• Depositional Landforms:

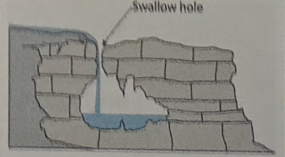
Landforms	Definitions	Pictures
Moraines	Deposition of unsorted glacial debris (till) along the sides of a glacier.	
Eskers	A long, sinuous ridge of sand and gravel deposited by melted glacial water streams flowing beneath the glacier.	
Outwash Plains	A gently sloping plain of well-sorted sand and gravel deposited by glacial meltwater beyond the glacier.	

Drumlins	An elongated, oval-shaped hill composed of glacial deposits (till), with its long axis parallel to ice flow. The upstream blunt end is called the stoss end and the downstream long & pointed end towards the flow of ice is called the tail or lee end. (Also known as "<i>Basket of eggs topography</i>").	 
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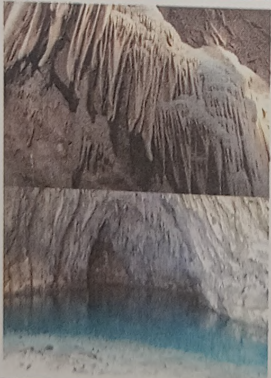
Landforms Created by Groundwater (also known as Karst Topography)

- The term "Karst" is derived from the Karst Plateau, which is located in south-western Slovenia and extends into north-eastern Italy.
- These are formed due to the solution (weathering) of soluble rocks like limestone and dolomite, which are rich in *calcium carbonate*.
- Since these rocks and such landforms are widely found in the Karst plateau region, all such groundwater landforms are called karst topography.
- Karst Topography location in India- Madhya Pradesh, Andhra Pradesh, Telangana, Odisha, and parts of the Himalayas.
- Erosional Landforms:**

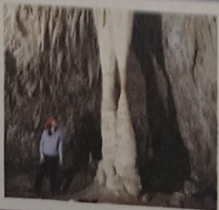


Landforms	Definitions	Pictures
Sinkholes	Depressions formed on the surface due to the solution or collapse of limestone, allowing water to drain underground.	
Swallow holes	Openings in limestone surface, through which streams disappear underground.	

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Dolines	Sinkhole enlarges due to continued solution of limestone by groundwater, forming a larger, circular or oval depression, called doline.	
Uvalas	A large, widespread, irregular depression formed by the joining or merging of several sinkholes or dolines is called uvala (<i>Sink valley</i>).	
Lapies	Irregular grooves and ridges on limestone surface formed due to the dissolving action of water.	
Limestone pavements	flat, widespread limestone surface broken into blocks due to solution along joints by groundwater.	
Caves	Underground hollows formed by the solution action of groundwater in limestone rocks (often containing stalactites and stalagmites - the depositional landforms).	

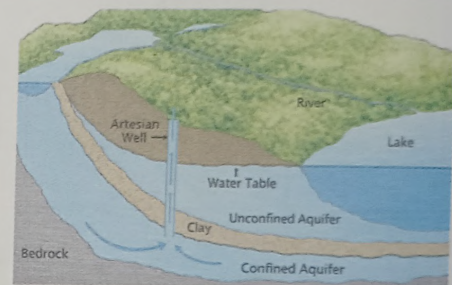
• Depositional Landforms:

Landforms	Definitions	Pictures
Pillars	A vertical structure formed when a stalactite and stalagmite join together.	

Stalactite	Elongated deposits of calcium carbonate hanging from the roof of caves, formed by dripping water. Hanging from the ceiling.	
Stalagmite	Cone-shaped deposits of calcium carbonate rising from the floor of caves, formed by accumulation of dripping water. Rising from the ground.	

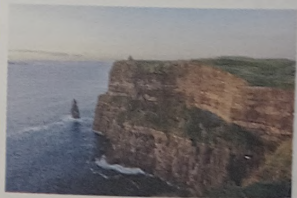
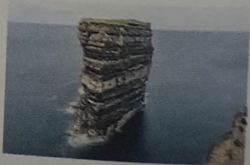
• Aquifer:

Groundwater stored between layers of hard rock below the water table.



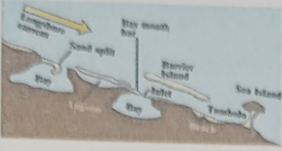
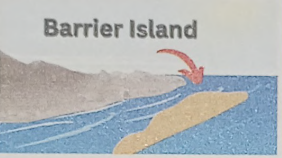
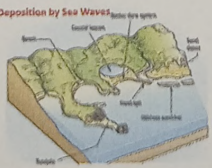
Landforms Created by Sea Waves

• Erosional Landforms:

Landforms	Definitions	Pictures
Sea Cliff	A steep rock face along the coast formed by continuous erosion by sea waves.	
Sea Caves	A hollow or cavity formed in coastal rocks due to continuous wave action along weak zones.	
Sea Arch	An arch-shaped opening formed when waves erode through a headland or cave.	
Sea Stack	An isolated, steep rock pillar in the sea formed when part of an arch collapses.	

LANDFORMS

• Depositional Landforms:

Landforms	Definitions	Pictures
Beaches	An accumulation of sand, pebbles, and other sediments along the shore deposited by sea waves.	
Spits	A narrow, elongated ridge of sand projecting into the sea, formed by longshore drift.	
Bars	A ridge of sand deposited across a bay or river mouth, often cutting it off from the sea.	
Barriers	A long, narrow offshore ridge of sand running parallel to the coast, formed by wave deposition.	
Lagoon	A shallow water body separated from the sea by a bar or spit.	
Tombolo	A ridge of sand or other deposits connecting an island to the mainland.	
Coastal sand dunes	Mounds or ridges of sand formed near the coast are called coastal dunes.	

Landforms Created by Wind (also known as Aeolian Landforms)

• Erosional Landforms:

Landforms	Definitions	Pictures
Yardang	Long, narrow ridges separated by grooves, formed due to wind erosion along the direction of prevailing winds.	

Mushroom rocks

A rock with a narrow base and broad top, formed due to faster erosion of lower soft rock layers than upper hard rock layers by wind (Also known as pedestal rocks).



Zeugen

Tabular rock masses with a flat top and steep sides, formed due to differential wind erosion of hard and soft rock layers.



Playa

A shallow, flat-floored depression in desert regions where water collects temporarily and dries up, leaving behind salts and fine sediments.



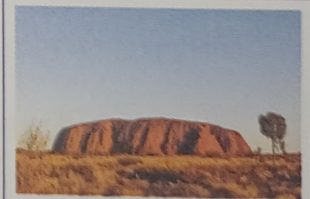
Piedplain

A broad, nearly level erosional surface formed after prolonged erosion in arid regions by the merging of several pediments.



Inselberg (Monadnock)

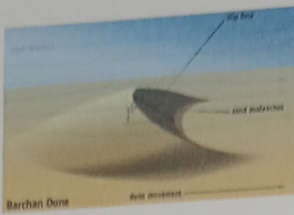
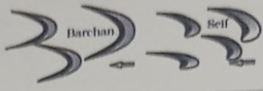
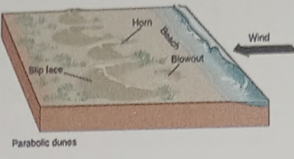
An isolated, steep-sided residual hill rising abruptly in a piedplain region.



• Depositional Landforms:

Landforms	Definitions	Pictures
Sand dunes/ Loess	When the grains of sand are very fine and light, the wind can carry it over a very long distance, such sand gets deposited in large areas and is called Loess or sand dunes.	

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Barchans	Crescent shaped dunes whose points or wings are directed away from wind direction. They are found in Jaisalmer, Rajasthan.	
Seif	Similar to Barchans with only one wing.	
Parabolic dunes	Sand dunes (also in crescent shape) form opposite to wind's direction, mainly because of the presence of vegetation.	
Transverse dunes	Sand dunes formed perpendicular to the direction of wind.	
Longitudinal dunes	Sand dunes formed parallel to the direction of wind.	
Oasis	Small fertile land with water source in arid region.	

ATMOSPHERE

- The atmosphere is a mixture of gases surrounding Earth, held in place by Earth's gravity.

Evolution of the Atmosphere

- Loss of Primordial Atmosphere:** Initially composed of hydrogen and helium, however this atmosphere was lost due to excessive solar flares.
- Volcanism:** The hot interior of Earth released gases through volcanic eruptions to form a new atmosphere (Degassing).
- Biological Modification:** Plants and living organisms played a key role in further transforming the atmosphere through Photosynthesis.

DIFFERENT LAYERS OF ATMOSPHERE

Troposphere → (Tropos: Region of mixing)

- Troposphere is the lowest layer of Earth's atmosphere.
- All weather phenomena (rain, storms, clouds, etc.) occur in this layer.

- Nitrogen (N_2): 78%
- Oxygen (O_2): 21%
- Argon (Ar): 0.9%
- Carbon dioxide (CO_2): 0.036%
- Neon (Ne): 0.0018%
- Helium (He): 0.00052%

Composition of Gases

Height variation

- At poles: 8 km
- At equator: 18 km
- Average: 13 km
- A **tropopause** is a line that separates Troposphere from Stratosphere.
- As we move upwards in the atmosphere, the density of the air decreases, and that is why the densest air is found in Troposphere.

Temperature at Tropopause

- At Equator: $-80^\circ C$
- At Poles: $-45^\circ C$

- Reason for temperature drop:** The atmosphere doesn't heat directly from solar radiation (insolation). Instead, the Earth's surface absorbs solar energy and re-emits it as long-wave infrared radiation (terrestrial radiation). This radiation warms the air near the surface.

As altitude increases, the air receives less terrestrial radiation, so the temperature drops with height.

- Negative Lapse Rate - Temperature increases with altitude.
- Positive Lapse Rate - Temperature decreases with Altitude.

Effects of Insolation

- At Equator: Insolation is less here, *due to presence of clouds*.
- At Tropics: Insolation is maximum here *due to minimal cloud covering*. Eg. Desert.

Factors influencing Insolation include:

- Transparency of the atmosphere
- Length of the day
- Earth's tilt and Rotation
- Latitude
- Altitude
- Position of Earth
- Surface reflection (Albedo)

Key Greenhouse Gases

- Greenhouse gases keep earth warm by trapping heat in the atmosphere.
- Key greenhouse gases: Water vapor, CO_2 , CH_4 (Methane), N_2O (Nitrous Oxide), O_3 (Ozone).

Heat Transfer Mechanism

- The transfer of heat in the atmosphere occurs through three primary mechanisms: **Conduction, Convection, and Advection**.
- Conduction:** The process by which heat is transferred directly from one molecule to another without actual movement of air.
- Convection:** The vertical transfer of heat through the movement of air masses.
- Advection:** The horizontal transfer of heat due to the movement of air. Eg. Loo is a result of advection.

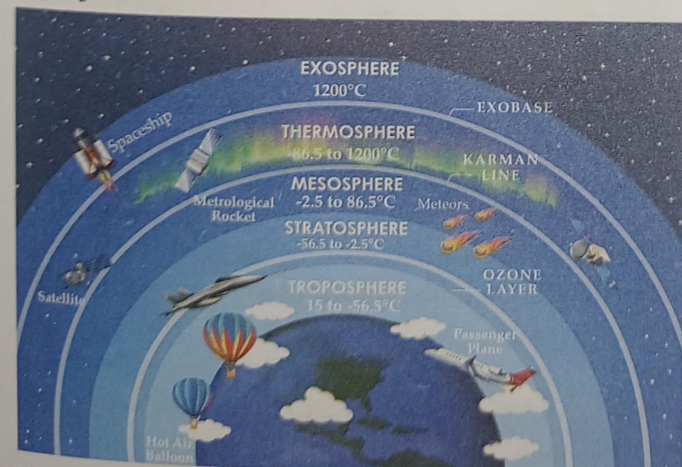
- NO_2 and SO_2 are not a Greenhouse gas.

Albedo

- Albedo refers to the proportion of incident light or radiation that is reflected by a surface.
- Highest albedo** is seen in **ice caps** or **glaciers**.

Albedo Values of Different Surfaces and Clouds

Earth surface	(%)	Clouds	(%)
Oceans, Lakes	8	Shallow broken clouds	30
Land Surfaces	14-18	Cu, Cl, Cs, Cc	35
Sand, Desert	27	St	40
Ice and Snow	50-70	Thick clouds (Cs)	74
Sea ice	35	Ac, As, Sc	68
Old snow	59	Cu	75
Fresh snow	80	Ns	85
Asphalt (lowest)	5-10	Cb	90



- Lapse rate** is the rate at which temperature decreases with altitude in the atmosphere.

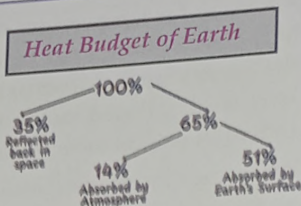
Standard Lapse Rate

- 165 meters → $1^\circ C$ decrease
- 1 km → $6.5^\circ C$ decrease

ATMOSPHERE

Heat Budget

• A heat budget is the balance between the heat that the Earth absorbs from the sun and the heat that it radiates back into space.



Temperature Inversion

- A phenomenon where temperature increases with height instead of decreasing.
- Temperature inversion leads to stability, means - No Rain.

Favourable Conditions for Temperature Inversion

1. **Long winter night:** During long nights, the Earth's surface loses heat rapidly through radiation, cooling the air near the surface.
2. **Still air:** Calm and stable air prevents the mixing of cold surface air with warmer air above, maintaining the inversion layer.
3. **Clear cloudless sky:** Clear skies enhance heat loss from the surface through terrestrial radiation, promoting rapid cooling and inversion formation.

Stratosphere (upto 50 km)

- This layer of atmosphere contains the **Ozone layer** (30-35 km above Earth's surface), which protects from harmful **UV rays** (B and C).
- While moving upwards, **temperature increases with altitude**, due to the absorption of ultraviolet (UV) radiation by ozone, which releases heat and creates a stable thermal inversion.
- Jet planes fly in this layer.
- **Stratopause** divides Stratosphere and Mesosphere.
- UV- A rays which are less harmful mostly pass from the Ozone.

• **Ozone Day** is celebrated on **16 September** to commemorate the adoption of the **Montreal Protocol** in 1987 in **Montreal, Canada**. This international agreement aimed to phase out **Chlorofluorocarbons (CFCs)**, which deplete the ozone layer and contribute to the formation of the ozone hole.

- Ozone layer thickness is measured in **Dobson unit**.
- **Kigali Amendment** was made to phase out **HFCs (Hydrofluorocarbons)**.

Mesosphere (upto 85 km)

- It is the **coldest** layer of atmosphere.
- **Meteors burn up in the mesosphere** because the air there is dense enough to compress and heat up in front of them, making their surfaces vaporize.
- While moving upwards, **temperature decreases with altitude**.

Thermosphere (between 80-400 km)

- It is the **hottest** layer of atmosphere.
- While moving upwards, **temperature increases with altitude**.

- It contains electrically charged particles known as **ions**, and hence, it is also known as the **ionosphere**.
- Radio waves transmitted from the earth are reflected back to the earth by this layer.

Karman Line

- It is an imaginary boundary located **100 kilometers (about 62 miles)** above sea level that **separates Earth's atmosphere from outer space**.

Exosphere

- It is the 5th and last layer of atmosphere. Mostly composed of **Hydrogen** and **Helium**.

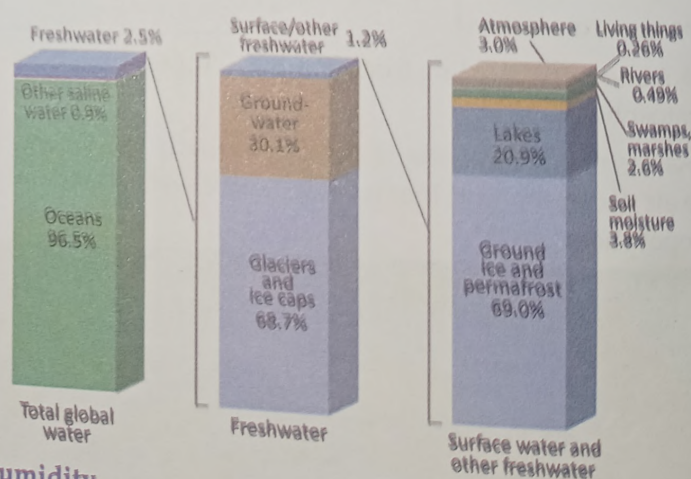
• **Isotherm Line:** These are lines or points on map/chart of the Earth's surface connecting points having same temperature at a given time.

WATER ON EARTH

- 97.2% of Earth's water is **saline**; only 2.8% is **freshwater**.
- Out of 2.8% freshwater: **Ice caps/glaciers** (2%), **Groundwater** (0.68%), **Lakes** (0.4%), **Atmosphere**, **River**.
- As a whole (Freshwater): **Ice caps/glaciers** (68.7%), **groundwater** (30.1%).

- 71% of Earth's surface covered with water
- 29% of Earth's surface is covered with land

Where is Earth's Water?



Humidity

- Humidity is the concentration of water vapour present in the air.

Types

1. **Absolute Humidity:** It is the **total mass of water vapor present in a unit volume** of air at a given temperature.
2. **Relative Humidity:** Actual moisture percentage compared to maximum capacity.

Absolute humidity is higher in warm air compared to cold air. This is because warm air can hold more water vapor due to its higher energy level.

Water Cycle

• Water cycle is the continuous *movement of water between the Earth's atmosphere and surface.*

Evaporation

• Evaporation is a process in the water cycle that *changes liquid water into water vapour.*

Latent Heat of Vaporization

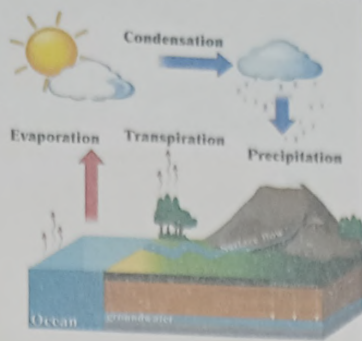
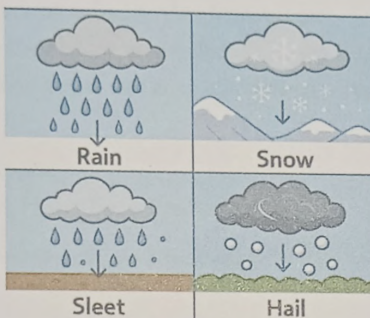
• The amount of heat energy required to change a unit mass of a liquid into a gas at a constant temperature is called latent heat of vaporization.

Precipitation

• Precipitation is any *liquid or frozen water* that forms in the atmosphere and falls back to the earth *in the form of rain, snow or hail.*

Types of Precipitation

1. **Rain:** Liquid water droplet that fall when the air temperature is above freezing point.
2. **Snow:** Frozen water crystals that are formed when temperature is below freezing point.
3. **Sleet:** Small pellets of ice that forms when raindrops freeze before reaching the surface.
4. **Hail:** It is a type of precipitation that falls to the ground in the form of ice balls or lumps. Concentric rings are seen in Hailstone.



• **Frost:** It forms when water vapor (gas) directly turns into solid ice crystals without becoming liquid first.

• **Radiation fog:** Forms when the ground cools overnight and chills the air above it. Eg. Valleys and low-lying areas.

• **Advection fog:** Forms when warm, moist air moves over a cooler surface. Eg. Coastal regions and plains.

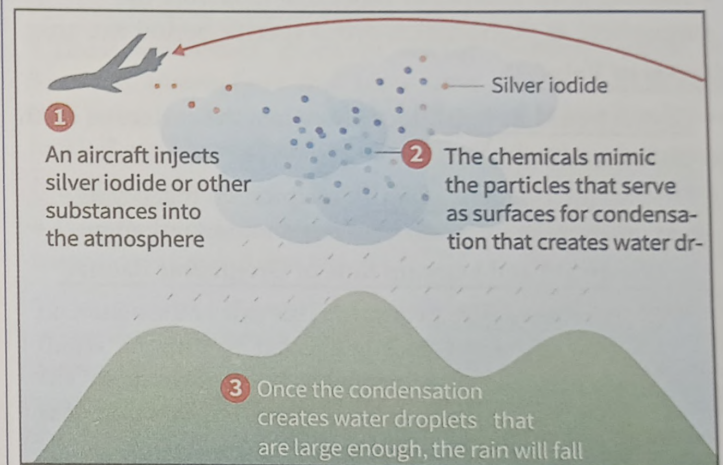
• **Smog:** Smog is a type of air pollution that results from the combination of smoke and fog.

Dew Point

- The dew point is the temperature at which air becomes fully saturated with moisture, *causing water vapor to condense into liquid droplets* and it deposits on the surface.
- In winter air cools down due to which *dew point reduces*, water vapour condenses into water.

Cloud seeding

Cloud seeding is a weather modification technique that involves dispersing substances like silver iodide, dry ice, or salts into clouds to increase the amount of precipitation that falls, such as rain or snow.



Condensation

• It is the process where water vapour (gas) becomes water (liquid).

Different forms:

• **Dew:** It is the moisture that form as a result of condensation.

• **Fog:** In winter, due to temperature inversion, cold air (which has low moisture-holding capacity) becomes saturated, and water vapour condenses near the ground, forming a low-level cloud called fog and has visibility *less than 1 km.*

In fog and mist, water vapor condenses around *hygroscopic nuclei* (aerosols like dust and smoke particles) instead of requiring a solid surface

• **Mist:** It is a phenomenon caused by small droplets of water suspended in the cold air, usually by condensation. Mist contains more moisture than fog, and has visibility *more than 1km.*

CLOUDS

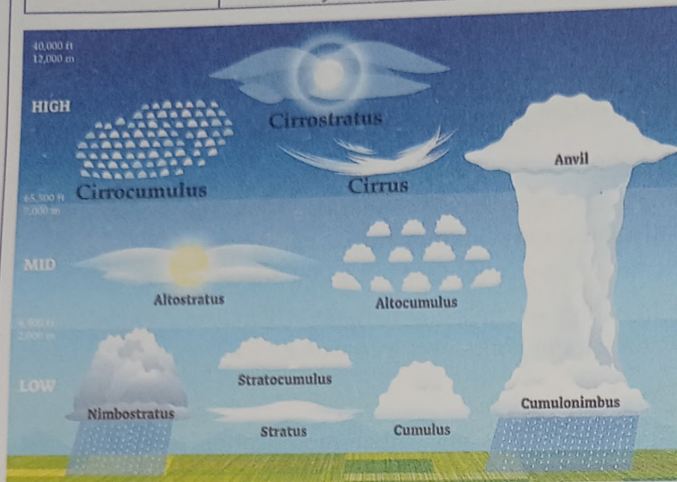
• Clouds are visible masses of tiny water droplets or ice crystals suspended in the Earth's atmosphere. Clouds formed through the collision-coalescence process.

• **Isoneph:** Line drawn on map, connecting the points containing same amount of clouds.

Different Types of Clouds

Clouds	Features	Height
Nimbus	Rain bearing; Opaque; Grey	Low-level
Stratus	Layered clouds	Low-level
Cumulus	Wool/Cauliflower shaped; flat base	4000-7000 m Middle level

Cirrus	Feather shaped clouds, do not cause rain	8000-12000 m High altitude
Cumulonimbus (anvil top)	Produce stormy weather that contribute to cyclones	Extensive vertical development



Rainfall

- Rainfall is the amount of water that falls from the atmosphere as precipitation over a specific period and area.

Types of Rainfall

- Convictional Rainfall** It occurs when the surface of earth is heated up by the Sun.
- Orographic / Relief Rainfall** Precipitation that occurs when moist air is forced to rise over a mountain range.

Windward and Leeward Side of Orographic Rainfall

- Windward Side (wet side):** This side of the mountain facing the wind (mostly affected by Anabatic wind) receives the heavy rainfall as the air rises and cools there. Eg. *Western sides of Western Ghats (Mumbai (Maharashtra) and Mangalore (Karnataka))*.
- Leeward Side (dry side):** This side of the mountain faces away from the wind (mostly affected by Katabatic wind) receives less rainfall because the air loses its moisture on the windward side before descending dry. It is also called a **rain-shadow region**. Eg. *Eastern side of Western Ghats (Vidarbha (Maharashtra) and Bangalore (Karnataka))*.

- Frontal/Cyclonic Rainfall:** It is a type of precipitation that occurs *when warm, moist air meets cold, dry air mass*.

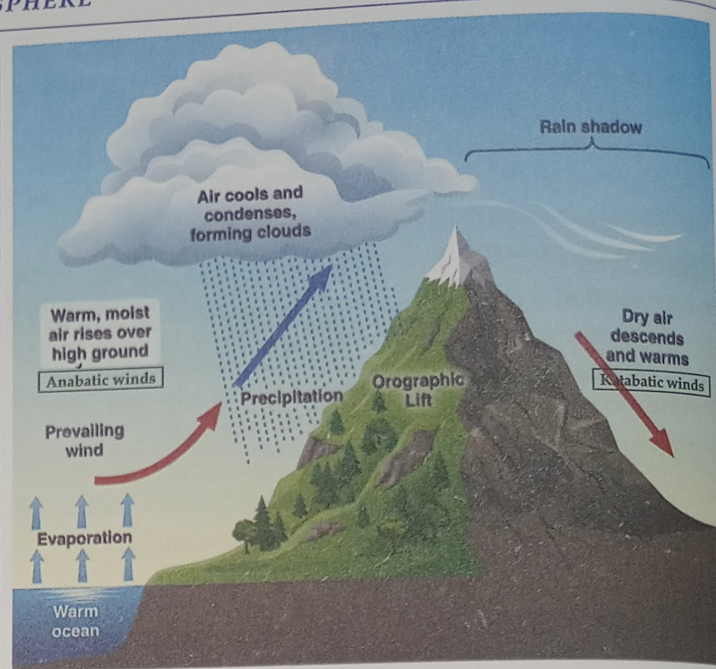
Convictional rainfall

Orographic rainfall

Cyclonic rainfall



Types of Rainfall



Windward and Leeward Side of Mountain

WIND AND PRESSURE DIFFERENCE

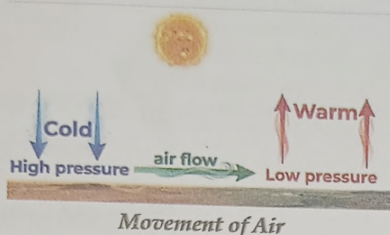
- **Wind** blows because of *differences in air pressure* in the Earth's atmosphere. Air moves from high pressure to low pressure. The sun heats the Earth unevenly, so some places get warmer than others. At the Equator, there is generally low air pressure due to the rising of warm and light air; while at the Poles, there is high air pressure because of the sinking of cold and dense air.

A wind is named based on the direction from which it originates.

Types of Wind

- Local Winds
- Permanent Winds

1. **Local Winds:** Local winds are winds that blow over short distances and are typically caused by *local geographic features and temperature differences* in specific areas.



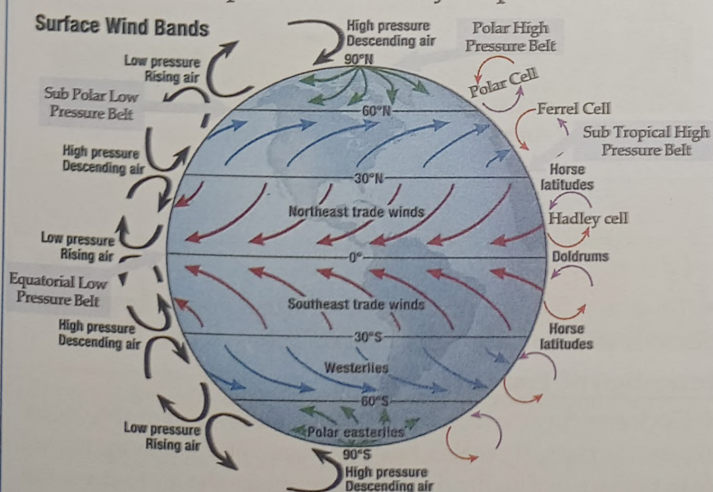
2. **Permanent Winds:** Permanent winds, also known as planetary winds, are winds that blow consistently from the same direction throughout the year, moving air across large areas of earth.

Types of Permanent Winds:

- (a) Trade Winds —
 - Northern Easterlies
 - Southern Easterlies
- (b) Westerlies
- (c) Polar Easterlies

Trade Winds:

Trade winds blow from subtropical high-pressure areas to the equatorial low-pressure belt. They are permanent winds.



Easterlies and Westerlies

- They are steady winds that blow from east to west near the Earth's equator.

Location

- They are found between approximately 30° latitude (north and south) and the equator.
- They blow from northeast in the Northern Hemisphere and southeast in the Southern Hemisphere towards the equator.

Cause

- They are created due to *Coriolis effect (rotation of the Earth)* and the unequal heating of the Earth's surface.

Westerlies

- Westerlies are winds that blow from *west to east* in the mid-latitudes.

When the barometric readings suddenly decrease, it indicates that the weather will be very stormy.

Location

- They blow between 30° and 60° latitude (in both hemispheres).
- They come from the southwest in the Northern Hemisphere and the northwest in the Southern Hemisphere.

Cause

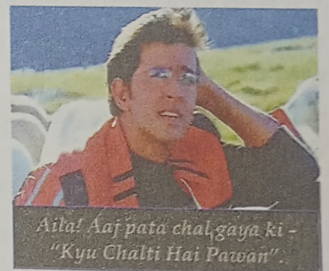
- The *Coriolis effect* causes these winds to deflect towards the east.

Roaring Forties, Furious Fifties, Shrieking Sixties

- These are names given to strong Westerly winds in Southern Hemisphere that blow over the 40°S, 50°S, and 60°S latitudes. Reason for their high speed is less landmass in Southern Hemisphere.

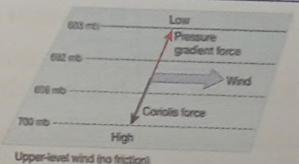
Coriolis Force

- Coriolis force is an apparent force that is caused due to *rotation of Earth*.
- The Coriolis force is responsible for deflecting winds towards the *right in the Northern Hemisphere* and towards the *left in the Southern Hemisphere*.
- Coriolis force is *maximum at the Poles* and *zero at the Equator*.
- The Coriolis force is directly *dependent on the wind speed*. The faster the wind moves, the stronger the Coriolis force becomes and greater is the wind deflection.



GEOSTROPHIC WIND

- Geostrophic wind is the wind that blows when two forces, the pressure gradient force and Earth's coriolis force, balance each other out.



- The wind *moves parallel to isobars* (isobars are lines on a weather map that connect areas with the same air pressure) or *perpendicular to Pressure Gradient Force*, without changing direction, usually higher up in the atmosphere where there's less friction.

ITCZ, Doldrums, and Horse Latitude

- Full form: **Intertropical Convergence Zone**
- The ITCZ forms where the **northeast trade winds** from the Northern Hemisphere meet the **southeast trade winds** from the Southern Hemisphere.
- It is typically found between the **Tropic of Cancer and Tropic of Capricorn**. In the month of July it is located at 20°N-25°N.

• Wind direction is measured using: **Wind vane**
• Wind speed is measured using: **Anemometer**
• Humidity is measured using: **Hygrometer**

Movement of ITCZ

- The ITCZ **moves seasonally**, following the **sun's position**.
- In the Northern Hemisphere, it moves north in the summer and south in the winter.
- The ITCZ moves more north and south over land than over water because the ocean heats up more slowly as compared to land.

Doldrums

- The Doldrums is a **low pressure area** around the Equator where the **prevailing winds are calm**.
- They are found between **10°N and 10°S**.
- The Doldrums **overlap with the ITCZ region**.

Horse Latitude

- Horse latitudes are regions located about **30°N and 30°S of the equator**.
- It is an area of subtropical high.
- It is a high-pressure area at divergence of trade winds and westerlies.
- They are characterized by Sunny skies, calm winds, and a very little precipitation.

LOCAL WINDS

Types of Local Winds

Winds	Warm/Cold Wind	Place where it Blows
Chinook	It is warm, dry wind and is also known as Snow eaters because it melt snow cover.	Canada, eastern side of Rockies
Sirocco	It is warm, hot and dry wind. It also causes "blood rain".	North Africa
Foehn	It is warm and dry wind.	Southern Alps mountain
Khamsin	It is hot, dry and dusty wind.	Egypt
Harmattan	They are dry, dusty warm winds. These winds are also known as "Dr. Wind".	Gulf of Guinea
Brickfielder	It is hot, dry and dusty wind.	Australia

Blizzard	They are very cold and carry snow.	Canada & Siberia
Mistral	These are strong, cold and dry winds.	France
Bora	These are strong and cold winds.	Adriatic Sea
Pampero	These are cold winds.	Pampas region, Argentina



Warm and Cold Winds

Land Breeze and Sea Breeze

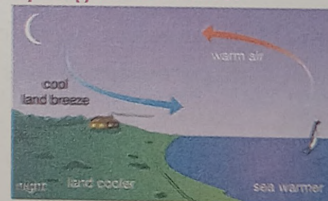
- Land Breeze and Sea Breeze are types of local winds that are due to temperature difference between land and water.

Land Breeze

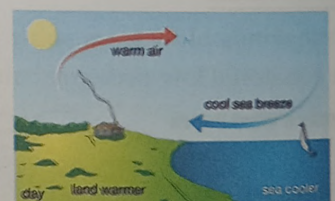
- It occurs at night when land cools faster than sea, creating high-pressure over land and low pressure over the sea, causing air to move from **land to sea**; more common in **autumn and winter** (land cools down fast because of its low specific heat).

Sea Breeze

- It occurs during the day when land heats up faster than the sea, creating low pressure over land and high-pressure over the sea, causing air to move from **sea to land**; more common in **spring and summer**.



Land Breeze



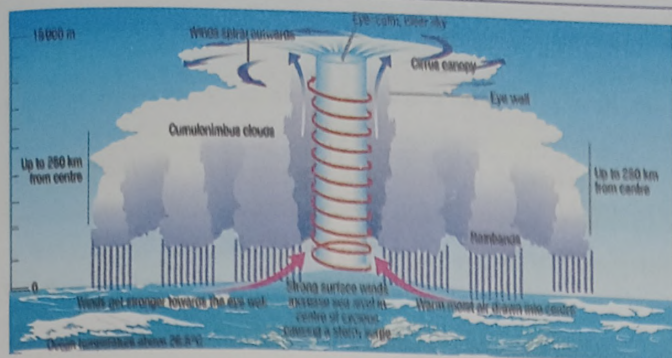
Sea Breeze

CYCLONE

- **Cyclones are rotating mass of air.**
- They are large, powerful storm that forms over warm ocean waters, with strong winds and heavy rain.

Formation of Tropical Cyclones

- Cyclones form when warm, moist air rises over the ocean. As the air rises, it creates low pressure, and cooler air rushes in to replace it. The Earth's rotation causes the air to spin, and the storm grows stronger. When the storm gets big enough, it becomes a cyclone with strong winds and heavy rain.
- A cyclone rotates **anti-clockwise** in **Northern Hemisphere** and **clockwise** in **Southern Hemisphere**.



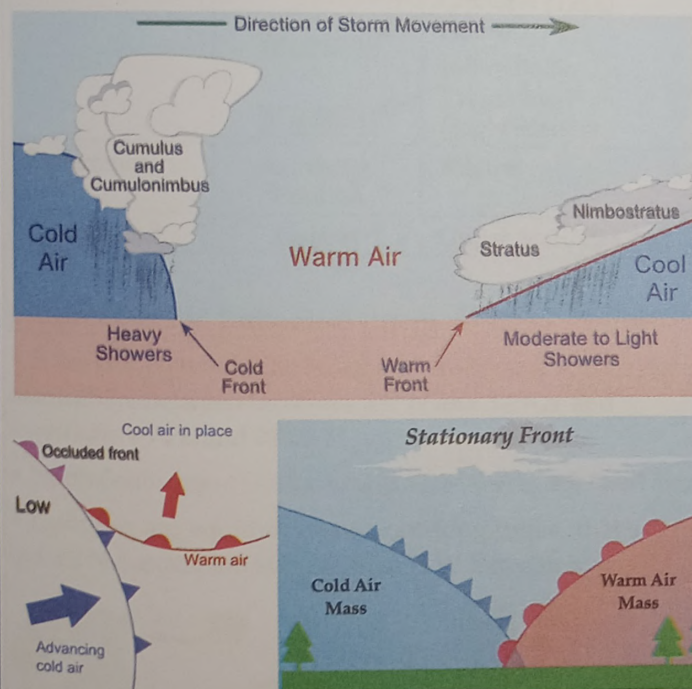
Favorable Conditions for Formation of Tropical Cyclone

- Large sea surface temperature (27°C)
- Atmospheric Instability
- Coriolis Force
- Pre-existing weak low pressure area

During cyclone, Cumulonimbus clouds are formed with extensive vertical development.

Formation of Temperate Cyclone:

- Mid-latitude cyclones or temperate cyclones or extra-tropical cyclones occur due to frontogenesis.
- They originate due to convergence of polar easterly cold air mass and sub-tropical westerly warm air mass in sub polar low pressure belts around 60° to 65° latitudes.
- Since the air masses are of two different properties, they can't intermingle readily. They mix with each other very slowly. As a result, a front develops between them.



Frontogenesis: It involves convergence of two distinct air masses.

Frontolysis: It involves Weakening or Destruction of a Front.

Different Names of Cyclones around the World

- Hurricane: Atlantic Ocean
- Typhoon: South China Sea and Western Pacific Ocean

- Willy-Willy: Australia
- Cyclone: India

Cyclone names are decided by regional committees, like the WMO/ESCAP Panel on Tropical Cyclones for the Bay of Bengal and Arabian Sea, that use pre-approved, sequentially assigned lists of names which are given by total 13 countries.

- Bay of Bengal (BOB) experiences more Cyclones than Arabian Sea because of several reasons

- It is surrounded by landmass which leads to more heat transfer in BOB.
- It is Funnel shaped, hence wind rushes in BOB from all directions.
- More Fresh water Influx.
- Monsoonal trough.

Tropical and Temperate Cyclone (Extra-Tropical Cyclone)

Tropical Cyclone	Temperate Cyclone
They originate over sea	They originate over sea and land
They are formed due to certain favourable condition over the sea, as mentioned above	They are formed due to frontogenesis i.e. where the warm and cold air mass meet each other
They cover less area but cause major destruction	They cover large area but cause less destruction
They occur rarely	They occur frequently
They move from east to west	They move from west to east

Cyclones form in every ocean except - South Atlantic Ocean & South Eastern Pacific Ocean.
Cyclones do not form in the Equator as there is a lack of Coriolis Force at Equator.

Anticyclone

- An anticyclone is formed around an **area of high pressure** where the air sinks, causing clear skies and calm weather. It usually brings dry, sunny conditions.
- An anticyclone rotates clockwise in Northern Hemisphere and anti-clockwise in Southern Hemisphere.

OCEAN CURRENTS

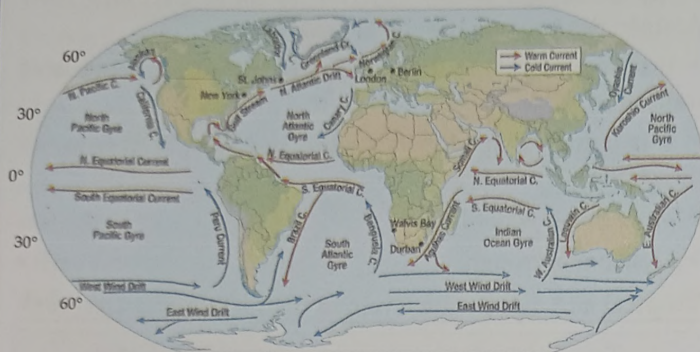
- Ocean currents are large movements of water in the ocean that flow in a specific direction.
- Ocean water moves in two directions: **horizontally and vertically**.
- **Horizontal movements** are referred to as **currents**, while **vertical changes** are called **upwellings** or **downwellings**.

Thermohaline Circulation: It is a global ocean current system driven by differences in seawater temperature and salinity, which affect its density. This process, often called the "global conveyor belt".

Reasons for Movement of Ocean Currents

1. Wind
2. Density difference
3. Gravity
4. Coriolis force
5. Coast line

- Cold water moves from Poles to Equator.
- Warm water moves from Equator to Poles.



Warm and Cold Ocean Currents Warm Currents

TRICK TO REMEMBERCold Ocean Currents → (BP GOLF World Cup California)

B	P	G	O	L
Benguela	Peru/Humbolt	Greenland	Oyashio	Labrador
F	World	Cup	California	
Falkland	West Australian Drift	Canary	California	

Warm Ocean Currents

- Brazil, Mozambique, Agulhas, Gulf Stream, Alaska, Norwegian, Kuroshio and E. Australia.

Effects of Ocean Currents

1. Cold ocean currents: They largely contribute to the formation of *coastal deserts in west coast regions of tropical and sub-tropical continents*. This is because the cold water makes the air above it cooler and less able to hold moisture.
2. Warm ocean currents + Cold ocean currents: Together they create *best fishing zones due to deposition of planktons*. The meeting of warm and cold ocean currents also *creates fog that is dangerous for harbouring*.



Desert around the world

Thermocline is the point below which temperature declines suddenly.

Halocline is the point below which salinity increases rapidly in sea.

Gyre: A system of ocean currents rotating in a circular pattern, by coriolis force.

INDIA

Some Facts About India

- India was a part of Gondwana land and is situated in continent of Asia.
- In terms of area: **7th** (India occupies **2.4%** of the world's area).
- In terms of population: 1st (**17.7%** of world's population lives in India).

Rank of countries around the world in terms of area

- Russia
- Canada
- China
- USA
- Brazil
- Australia
- India
- Argentina

- India has 28 States and 8 Union Territories

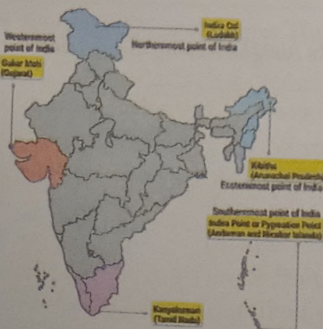
Area	State	UT
Largest	Rajasthan	Ladakh
Smallest	Goa	Lakshadweep
Population		
Largest	Uttar Pradesh	Delhi
Smallest	Sikkim	Lakshadweep

Extreme Points in India

Extremes	State	Point
Northernmost	Himachal Pradesh	Indira Col (Ladakh)
Southernmost	Tamil Nadu	Kanyakumari/Cape Comorin (Tamil Nadu)
		Indira Point/Pygmalion Point (A&N islands)
Easternmost	Arunachal Pradesh	Kibithu
Westernmost	Gujarat	Guhar Moti/Sir Creek

Longitudinal and Latitudinal Extent

- It lies completely in the Northern hemisphere and Eastern hemisphere between **latitudes $8^{\circ}4'N$ and $37^{\circ}6'N$** and **longitudes $68^{\circ}7'E$ and $97^{\circ}25'E$** .
- North-South extent (**3214 km**) and East-West extent (**2933 km**).
- Latitudes are equidistant but not longitudes, that's why India's N-S extension is more than E-W Extension.



Extreme Points of India;



Longitudinal and Latitudinal Extent of India

Topic of Cancer

- India is divided by Tropic of Cancer **$23^{\circ}30'N$** in almost **two equal parts**.
- The Tropic of Cancer passes through **8 States** out of which Jaipur (Rajasthan), Agartala (Tripura), Aizawl (Mizoram) are only capitals that are above the tropic of Cancer.

TRICK TO REMEMBER → (Gujarati Raja Made Chief Justice Win The Meeting)

Gujarati	Raja	Made	Chief
Gujarat	Rajasthan	Madhya Pradesh	Chhattisgarh
Justice	Win	The	Meeting
Jharkhand	West Bengal	Tripura	Mizoram

Indian Standard Time

- The difference of time between **Gujarat** and **Arunachal Pradesh** is **2 hours**; but both their clocks show the same timing. This is because India follows a standard time. **Indian Standard Time** (IST is 5:30 hrs ahead of GMT) is along **longitude $82.30^{\circ}E$** . It passes through the city of **Mirzapur** in **Uttar Pradesh**.

- The IST passes through 5 States:**

TRICK TO REMEMBER → (MOUCA)

M	O	U	C
Madhya Pradesh	Odisha	Uttar Pradesh	Chhattisgarh
A			
Andhra Pradesh			

- Time Zones:** France (13), Russia (11), USA (6) → because of many territories).
- Tropic of Cancer meets IST at:** **Korea district, Chhattisgarh (Baikunthpur)**.



Indian Standard Time of India

LAND BORDER OF INDIA

India's International Boundary

Countries that share boundary with India:

TRICK TO REMEMBER → (Ba CH PaN में MBA)

B	Ch	Pa
Bangladesh (4096.7 km)	China (3488 km)	Pakistan (3323 km)
N	में	M
Nepal (1751 km)	Myanmar (1643 km)	B
A		
Afghanistan (106 km)		

Total Land Border Area: 15,106.7 km

India and Bangladesh Border Shared By

TRICK TO REMEMBER → (MAM Take Water)

M A M Take Water
Meghalaya Assam Mizoram Tripura West Bengal

It is surrounded by **Bangladesh** on three sides

- The **Radcliffe Line** is the boundary that separates India and Bangladesh.

India and Nepal Border Shared By

TRICK TO REMEMBER → (SUBBU)

S U B B U
Sikkim Uttar Pradesh Bihar West Bengal Uttarakhand

- Treaty of Sugauli** was signed in **1816**, that established the boundary line between India and Nepal.

India and Bhutan Border Shared By

TRICK TO REMEMBER → (BASA)

B A S A
West Bengal Assam Sikkim Arunachal Pradesh

India and Pakistan Border Shared By

TRICK TO REMEMBER → (GRP J&K Ladakh)

G R P J&K Ladakh
Gujarat Rajasthan Punjab Jammu and Kashmir Ladakh

- The **Radcliffe Line** (**Sir Cyril Radcliffe drew the line**) is the boundary that separates India and Pakistan.
- The **Line of Control (LoC)** is a military control line that separates India and Pakistan in the disputed region of J&K.

India and China Border Shared By

- J&K, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh.

- Indo-China Border is divided into three sectors:**

1. **Western sector (Johnson Line)** defines the boundary between **India** and **China** in the Western sector).

2. **Middle sector.**

3. **Eastern sector (McMahon Line)** defines the boundary between **India** and **China** in the Eastern sector).

India has consistently maintained that a region, also known as the Trans-Karakoram Tract, lying north of the Siachen Glacier in Ladakh, historically part of the erstwhile princely state of Jammu and Kashmir, was illegally ceded to China by Pakistan under the 1963 China-Pakistan Boundary Agreement. Which region is being referred to Shaksgam Valley.

- The Galwan River Valley, which is located in north-eastern Ladakh, east of the Aksai Chin plateau, is also illegally occupied by China.

- The **Line of Actual Control (LAC)** is a military control line that separates Indian areas of **Ladakh** from **Aksai Chin**.



LOC and LAC



Indo-China Border

Vibrant Villages Programme

- It is a Centrally sponsored Scheme announced in Union Budget 2022-2023 (to 2025-2026) for development of villages on India-China border.

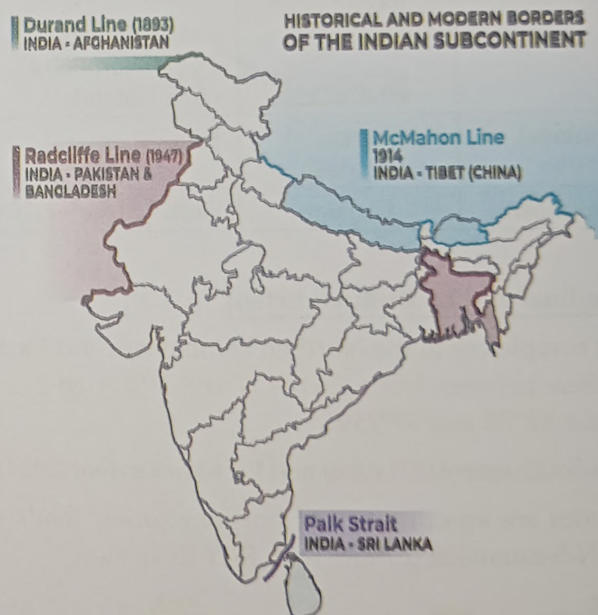
India and Myanmar Border Shared By

TRICK TO REMEMBER → (AruNa MaMi)

Aru Na Ma Mi
Arunachal Pradesh Nagaland Manipur Mizoram

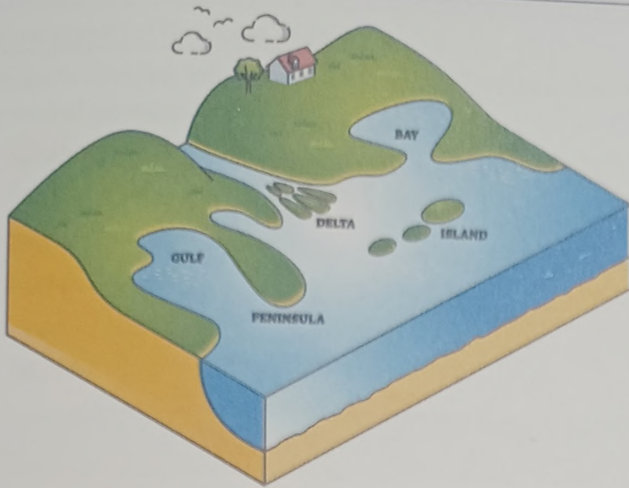
India and Afghanistan

- India and Afghanistan is separated by Union Territory: **Ladakh**.
- The border between India and Afghanistan is called the **Durand Line**.



Dividing Lines of India with its Neighbouring Countries

- Gulf:** A gulf is a portion of the ocean that penetrates land. They are generally **larger and more deeply indented** than **bays**. Gulfs are sometimes connected to the ocean by narrow passages of water called straits.
- Largest Gulf in the world:** Gulf of Mexico.
- Bay:** A bay is a body of water that is **partly surrounded by land**. Bays **empty out** into larger bodies of water like **oceans and lakes**.
- Largest Bay in the world:** Bay of Bengal.



Some Important Facts,

- **State sharing boundary with maximum number of states:** Uttar Pradesh (with 8 States 1 UT (Delhi)).
- **State sharing border with only one state:** Sikkim (with West Bengal) and Meghalaya (with Assam).
- **States sharing border with three countries:**
 1. **Sikkim** (with Nepal, Bhutan, China).
 2. **Arunachal Pradesh** (with Bhutan, China, Myanmar).
 3. **West Bengal** (with Nepal, Bhutan, Bangladesh).
- **One UT that shares boundary with three countries:** **Ladakh** (with Pakistan, Afghanistan, and China).

Few Organisations

Organisations under Central Armed Police Force (CAPF)

- **Indo-Tibetan Border Police (1962):** ITBP (under Ministry of Home Affairs) guards the country's border with China along the Indo-Tibetan region.
- **Sashastra Seema Bal (1963):** SSB (under Ministry of Home Affairs) guards the country's borders with Nepal and Bhutan.
- **Border Security Force (1965):** BSF is known as "India's first line of Defence" and was established after Indo-Pak War. It guards country's border with Pakistan and Bangladesh..
- **Assam Rifles:** It is under Ministry of Home Affairs but the operation control is under Ministry of Defence.
- **Central Reserve Police Force (CRPF):** It is India's largest paramilitary force and is under Ministry of Home Affairs.

Indian Armed Forces (IAF)

- **Air Force/Navy/Army:** They are under **Ministry of Defence**.

Naxalite affected areas are called as "Red Corridors"

COASTAL BOUNDARY OF INDIA

- India's total coastal boundary length is approximately **11,098.81 km**.

This includes:

- **Mainland coastline: 7870.51 km.**
- **Island territories: 3,228.30 km.**

- India's coastline touches the Arabian Sea, Bay of Bengal, and the Indian Ocean, spanning across 9 coastal states (Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Odisha, West Bengal) and 2 island groups (Andaman & Nicobar Islands and Lakshadweep Islands).

Length of Coastline of India

The State-wise/UT-wise "Length of Coastline of India" are here by mentioned below.

S.No.	State/Union Territory	Coastline length (in Km)
1.	Gujarat	2,340.62
2.	Maharashtra	877.97
3.	Goa	193.95
4.	Karnataka	343.30
5.	Kerala	600.15
6.	Tamil Nadu	1,068.69
7.	Andhra Pradesh	1,053.07
8.	Odisha	574.71
9.	West Bengal	721.02
10.	Daman & Diu	54.38
11.	Pondicherry	42.65
12.	Lakshadweep Islands	144.80
13.	Andaman & Nicobar Islands	3,083.50
Total Length of Coastline of India		11,098.81

Longest coastline of India:

- **UT: Andaman & Nicobar Islands** → 3,083.50 km
- **State: Gujarat** → 2,340.62 km
- **Shortest coastline of India: Goa** → 193.95 km

States with longest coastline

1. **G: Gujarat**
2. **T: Tamil Nadu**
3. **A: Andhra Pradesh**

TRICK
GTA

Largest coastal boundary of the world: Canada

Union Territories of India that share coastal boundaries are:

1. **Andaman and Nicobar Islands** (Bay of Bengal) → located on eastern coast of India.
2. **Lakshadweep** (Arabian Sea).
3. **Puducherry** (Bay of Bengal).
4. **Daman and Diu** (Arabian Sea).

Sea Borders of India

- India shares its sea borders with two countries: **Sri Lanka and Maldives**.
- India and Sri Lanka is separated by **Palk Strait** (a narrow water body, which connects two large water bodies or separates two landmasses).

- India's sea borders include the *Bay of Bengal to the east, the Arabian Sea to the west, and the Indian Ocean to the south.*
- *Gulf of Mannar* is a shallow sea between *India and Sri Lanka*, located near the southern tip of India.

United Nations Convention on the Law of the Sea (UNCLOS)

- It was *adopted on 1982* but came into *force on 1994*.
- It is an international treaty that establishes rules and guidelines for the use and protection of the world's oceans.
- *It covers 5 zones: Internal Waters, Territorial Sea, Contiguous Zone, Exclusive Economic Zone (EEZ) and High Seas.*

Territorial Limit of India

- The length of coastline is measured in *Nautical Miles (nm)*.
1 nautical miles 1.852 km.
- *India's territorial limit extend upto 12 nm (12 nm = 22 km) from the coast.*
- **Contiguous Zone:** A contiguous zone is a maritime area that lies beyond a country's territorial sea. It extends upto *24 nm*.
- **Exclusive Economic Zone:** It is a maritime area that extends upto 200 nm from a coastal nation's territorial sea.

Kanyakumari is a place in India where the Bay of Bengal, the Indian Ocean, and the Arabian Sea meet.

New Pamban Bridge, the first Vertical lift sea Bridge of India which has been inaugurated recently connects Pamban (Rameswaram island) to Mandapam (Mainland India).

**PLEASE NOTE THAT

- Recently Government of India on 29th April 2025 released that the length of India's coastline has been recalculated and is now 11098.81 kms.



Physiographic Divisions of India

• India is divided into six physiographic divisions based on the varied physical features.

1. *Himalayas/Northern Mountains*
2. *Northern Plains*
3. *Peninsular Plateau*
4. *Indian Desert*
5. *Coastal Plains*
6. *Islands*

MOUNTAINS

• Mountains are landforms that are much higher than the surrounding land. They are usually steeper than hills and often have a sharp peak, or summit.

Types of Mountains

1. **Fold Mountains:** They are formed when tectonic plates collide, causing the Earth's crust to buckle and fold. These mountains are often characterized by parallel ridges and valleys.

Types

1. **Young Fold Mountains:** Mountains that were formed by tectonic activity a few million years ago.
Eg. *Himalayas, Andes (located in South America and is world's longest mountain range), Alps (Europe), Rockies (North America), Atlas (Africa), Caucasus mountains.*
2. **Old Fold Mountains:** Mountains that were formed by the folding of the Earth's crust in the distant past.
Eg. *Aravalli (residual mountains) Ural (separates Asia and Europe) Appalachians (North America).*
3. **Block Mountains:** Block mountains are formed when large areas of the Earth's crust are broken and displaced due to vertical slippage.
Horst → *raised block of land bounded by faults*, often appearing as a ridge or plateau and *Graben (also known as Rift Valley)* *lowered block of land between two faults*, often forming a valley are formed in block mountains.
Eg. *Vindhya and Satpura, Vosges Mt. (France), Harz Mt. (Germany), Great African Rift Valley (world's longest Rift Valley).*
3. **Volcanic Mountains:** They are formed when magma from the Earth's mantle rises to the surface and erupts, creating lava flows that build up over time.
Eg. *Mt. Stromboli (known as "Lighthouse of Mediterranean", Mt. Cotopaxi (Ecuador), Fujiyama (Japan), Barren and Narcondam (Andaman & Nicobar Islands), Ojas del Salado (world's highest active volcano in Chile-Argentina border), Mt. Kilimanjaro (Africa).*

HIMALAYAS

• The Himalayas are one of the *youngest fold mountain* ranges in the world of tectonic origin and comprise mainly sedimentary rocks.

Geosyncline: It is an *obsolete geological concept* that described a long, linear trough in the Earth's crust where vast amounts of sediment accumulate, which was believed to later be deformed into mountain range.

Formation of Himalayas

• They were formed when the *Indo-Australian tectonic plate* and *Eurasian (Asian) tectonic plate collided*, this caused the closure of the ancient *Tethys Sea*, leading to the folding and uplifting of sediments deposited on its seabed. This resulted in creation of multiple massive mountain ranges which are collectively known as the Himalayas.

Four Mountain Ranges in Himalayas**1. Trans-Himalayas**

- It is located north of the Great Himalayas, this region is characterized by high plateaus and arid landscapes.
- *Average elevation: 3000 m.*
- Kailash range (China) is also part of Trans Himalaya.

The prominent ranges under Trans Himalayas from North to South order**1. Karakoram Range**

- It forms India's boundary with *Afghanistan and China.*
- Highest peak of this range is *K2*, also known as *Mt. Godwin-Austen (8611 m)*, which is also *world's second highest peak.*
- Mt. Rakaposhi, Mt. Gasherbrum is also a part of Karakoram Range.

2. Ladakh Range

- It has the *highest slope.*
- *Shyok* (tributary of Indus) river flows between *Karakoram* and *Ladakh* range.

3. Zaskar Range

- *Kamet peak* is the *highest peak* of this range.
- Indus river flows between *Ladakh* and *Zaskar* range.

Glacier of Karakoram Range

- | | |
|--------------------------------------|-------------------------------|
| 1. Siachen | 2. Baltoro (Gilgit-Baltistan) |
| • 2nd longest non-polar glacier. | 3. Hispar |
| • Highest battlefield (5700 m above) | 4. Biafo |
| • Operation Meghdoot (1984) | |

*Longest non-polar glacier in the world: *Fedchenko Glacier* in *Tajikistan*

*Longest glacier in the world: *The Lambert Glacier* in *Antarctica*

2. Greater Himalayas

- They are also known as *Himadri/Inner Himalayas.*
- They are asymmetrical in nature.
- The core of Greater Himalayas is composed of *granitic rock.*